Explainable Longitudinal Depression Severity Prediction and Progression Tracking from Passive Wearable Sensing: A Subject-Independent Benchmark and Clinical Decision Support Framework

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Abstract

Major depressive disorder (MDD) is a leading cause of global disability, yet conventional clinical assessment remains episodic, reactive, and reliant on retrospective self-report instruments such as the Patient Health Questionnaire-9 (PHQ-9). While passive digital sensing via wearable actigraphy and smartphone telemetry offers continuous behavioral monitoring, previous machine learning studies frequently suffer from participant data leakage, class imbalance collapse on rare severe depression tiers, and an inability to forecast dynamic longitudinal symptom shifts. In this paper, we present an end-to-end, explainable machine learning and longitudinal sequence modeling framework for depression severity assessment and clinically meaningful progression tracking. Evaluated on a multi-modal cohort of 4,948 participants comprising 35,694 monthly tracking records and 10,866 labeled PHQ-9 assessments across 150 physiological, circadian, movement, demographic, and clinical features, our methodology enforces a strict, participant-disjoint partition (3,463 training, 742 validation, and 743 test participants; 1,575 test assessments). We evaluate tabular gradient boosted trees, regularized neural networks, and windowed recurrent Gated Recurrent Unit (GRU) architectures across multi-period observation horizons ($W \in \{1, 2, 3, \text{Full}\}$). Prospectively selected on validation Macro-F1 (0.3359), the single-period windowed model ($W=1$) demonstrates robust generalization across held-out test data (Test Macro-F1: 0.3204, Quadratic Weighted Kappa: 0.4851). In longitudinal transition analysis across 973 test assessment pairs, our models overcome the "Persistence Baseline Paradox"—wherein a naive persistence baseline achieves 76.46% raw accuracy by predicting chronic stability while exhibiting 0.0% directional agreement—achieving 38.33% to 44.60% directional agreement in capturing true clinical changes. To address severe-case under-detection, post-hoc threshold optimization tuned to $T_4^* = 0.2100$ increases severe-tier sensitivity from 25.0% to 41.7% (a 66.7% relative improvement). Finally, TreeSHAP and permutation importance establish that sleep and circadian disruption contribute 45.4% of total predictive attribution, followed by physical activity (33.5%) and clinical comorbidities (15.6%). These findings demonstrate that wearable behavioral telemetry can reliably support objective clinical triage when governed by strict subject-level evaluation and transparent decision auditing.

Keywords: Affective Computing, Digital Phenotyping, Longitudinal Sequence Modeling, Depression Progression, Minimal Clinically Important Difference (MCID), Gated Recurrent Units, Explainable AI, TreeSHAP.

1. Introduction

Major depressive disorder affects more than 280 million individuals worldwide, representing a primary contributor to the global burden of non-fatal disease and psychiatric disability [1], [3]. Despite its prevalence, standard clinical management continues to depend almost exclusively on intermittent outpatient consultations scheduled weeks or months apart. During these visits, symptom severity is predominantly quantified using retrospective psychometric rating scales, such as the Patient Health Questionnaire-9 (PHQ-9) [1] or the Hamilton Depression Rating Scale (HDRS). While standardized, these self-report instruments are fundamentally vulnerable to cognitive recall bias, transient affective states on the day of assessment, and social desirability pressures [4], [5]. Critically, episodic assessment fails to capture intra-individual symptom dynamics, sub-clinical deterioration, or the early warning signals of depressive relapse that manifest between scheduled appointments [6], [7].

Recent advances in digital phenotyping and consumer wearables—equipped with continuous tri-axial accelerometry, photoplethysmography, and smartphone telemetry—offer an unprecedented opportunity to observe passive behavioral and circadian markers continuously and unobtrusively in ecological daily life [4], [8]. Prior studies have demonstrated that parameters such as daily step volume, midday physical activity entropy, diurnal rest-activity ratios, and sleep fragmentation correlate with clinical depression severity [6], [8], [9]. However, translating these digital observations into robust, clinically trustworthy predictive models presents several severe methodological challenges:

1. **Pervasive Data Leakage in Affective Computing:** A substantial proportion of published digital sensing studies evaluate machine learning models using random row-level k-fold cross-validation or random observation-level splits. When multiple longitudinal records from the same individual appear in both training and test sets, models exploit idiosyncratic participant-specific baseline traits (such as unique stride cadence or baseline resting heart rate) rather than generalizable depressive biomarkers [9], [23]. This yields artificially inflated accuracies that catastrophically collapse when deployed on unseen patients.
2. **Extreme Ordinal Class Imbalance:** In real-world outpatient cohorts, depression severity exhibits severe right-skewed imbalance: the vast majority of tracking records correspond to minimal (Tier 0) or mild (Tier 1) symptoms, whereas moderately severe (Tier 3) and

severe (Tier 4) depression represent small fractions (<5%) of the population. Standard learning objectives (e.g., cross-entropy or accuracy maximization) incentivize algorithms to ignore rare high-risk tiers, resulting in dangerous clinical false negatives [12], [21].

3. **The Multi-Period Longitudinal Challenge:** While cross-sectional models map isolated feature vectors to static depression severity tiers, psychiatric treatment monitoring requires tracking *change over time*. Specifically, clinicians require answers to two questions: *Is the patient experiencing clinically meaningful improvement, stability, or worsening?* and *Does incorporating sequential behavioral history improve predictive power over current cross-sectional observations?*

4. **The "Persistence Baseline Accuracy Paradox":** In longitudinal psychiatric cohorts, symptom trajectories are predominantly stable across adjacent quarters. Consequently, a trivial "persistence baseline"—which naively predicts that a patient's score remains identical to their previous score—achieves deceptive, high raw classification accuracy while possessing exactly zero clinical utility for detecting deterioration or therapeutic response [2].

5. **Lack of Transparent Model Explanations:** Deep sequence models and complex ensemble architectures operate as black boxes. Clinicians cannot ethically act on uncalibrated, uninterpretable risk probabilities without an itemized audit trail identifying whether an alert is driven by fragmented sleep, physical inactivity, chronic pain, or socioeconomic hardship [14], [15].

To address these interconnected challenges, this paper formulates, implements, and evaluates an end-to-end framework for explainable longitudinal depression severity prediction and progression tracking. Grounded strictly in an audited multi-modal dataset of 4,948 participants, our study enforces participant-level split isolation, compares tabular and recurrent architectures, systematically evaluates multi-period historical observation windows, resolves the persistence paradox using Minimal Clinically Important Difference (MCID) metrics, and implements decision-support calibration and TreeSHAP explainability.

Research Questions

This investigation is structured around four primary research questions:

- **RQ1 (Participant-Independent Cross-Sectional Benchmark):** To what extent can passive movement, sleep, circadian, demographic, and clinical features predict ordinal PHQ-9 depression severity tiers when evaluated on strictly unseen participants?

- **RQ2 (Architectural Comparison Under Class Imbalance):** How do regularized neural networks, cost-sensitive gradient boosted trees, and recurrent sequence models compare when tasked with balancing overall discrimination against rare severe-class detection?

- **RQ3 (Value of Multi-Period Observation Horizons):** Does incorporating deeper sequential history ($W \in \{2, 3, \text{Full}\}$) improve predictive discrimination or directional change forecasting over a single-period observation window ($W=1$)?

- **RQ4 (Longitudinal Symptom Progression & Detection of Clinical Shifts):** Can learned predictive models outperform a persistence baseline in identifying non-zero, clinically meaningful symptom changes (≥ 5 PHQ-9 points), and what biological domains drive these predictions?

Contributions

The primary contributions of this paper are fourfold:

1. **Leakage-Free, Participant-Level Evaluation Protocol:** We benchmark tabular and sequential models across a disjoint cohort of 4,948 participants (3,463 train, 742 validation, 743 test) and 10,866 labeled assessments, proving that zero-leakage subject-level evaluation prevents the over-optimistic generalization claims pervasive in passive sensing literature.

2. **Systematic Multi-Window Longitudinal Recurrent Modeling:** We evaluate windowed Gated Recurrent Unit (GRU) architectures across controlled sequence horizons ($W=1, 2, 3, \text{Full}$), establishing that prospective model selection must be governed strictly by validation Macro-F1 to avoid post-hoc test snooping.

3. **Mathematical Resolution of the Longitudinal Persistence Paradox:** We demonstrate that while a persistence baseline achieves 76.46% MCID accuracy by predicting chronic stability for all cases, it achieves 0.0% directional agreement. Learned sequential models achieve 38.33% to 44.60% directional agreement, capturing true clinical shifts where naive baselines are completely degenerate.

4. **Clinical Decision Support via Calibrated Triage and TreeSHAP Auditing:** By tuning severe-class decision thresholds to $T_4^* = 0.2100$, we increase severe depression sensitivity by 66.7% relative to default argmax classification. We pair this with polynomial-time TreeSHAP attributions and permutation importance, providing itemized physiological receipts that quantify the dominant role of sleep-circadian fragmentation (45.4%) and actigraphy (33.5%) in automated mental health triage.

2. Related Work

2.1 Passive Sensing and Digital Biomarkers of Depression

The application of passive digital sensors to monitor affective disorders has expanded rapidly over the past decade [4], [5], [9]. Early exploratory studies by Saeb et al. [8] demonstrated that smartphone-derived GPS mobility metrics (e.g., location variance, circadian movement rhythmicity) correlated significantly with depressive symptom severity measured via the PHQ-9. Subsequent investigations incorporated wearable actigraphy to monitor sleep architecture, rest-activity cycles, and motor activity. Jacobson et al. [6] analyzed actigraphy-derived movement dynamics, demonstrating that circadian disruption and blunted diurnal activity patterns served as prominent digital biomarkers of major depressive episodes. Similarly, Cho et al. [7] tracked bipolar and depressive cohorts using commercial wristbands, demonstrating that light exposure, circadian phase shifts, and sleep duration fluctuations preceded mood episode transitions.

Despite these promising associations, systematic reviews by Cornet and Holden [5] and De Angel et al. [9] emphasized critical weaknesses across the digital sensing literature: small sample sizes (typically $N < 100$), short monitoring durations (<3 months), high missingness rates, and an overwhelming reliance on cross-sectional correlational analyses rather than prospectively validated predictive frameworks.

2.2 Longitudinal Sequence Modeling in Mental Health

Traditional machine learning approaches in psychiatry treat successive patient evaluations as independent, identically distributed (i.i.d.) observations [12]. However, psychiatric illness is inherently sequential, exhibiting complex autocorrelation, seasonal variation, and delayed treatment responses [6], [7]. Recurrent neural networks, particularly Gated Recurrent Units (GRUs) [10] and Long Short-Term Memory (LSTM) networks, are designed to model variable-length temporal sequences without suffering from vanishing gradients [11]. In clinical informatics, recurrent architectures have been deployed to predict intensive care mortality, sepsis onset, and chronic disease trajectory.

In mental health digital phenotyping, recurrent modeling remains underexplored due to the scarcity of long-term cohorts with repeated psychometric ground truth. While preliminary studies have applied RNNs to daily smartphone logs, they rarely systematically evaluate the depth of the historical observation window (W) or evaluate models on clinically meaningful symptom shifts defined by medical consensus [2], [9].

2.3 Imbalanced Learning and Threshold Optimization

Depression severity distributions in general and outpatient populations exhibit severe class imbalance [9], [21]. Standard machine learning classifiers trained via unweighted empirical risk minimization maximize overall accuracy by assigning the majority of instances to minimal or mild categories. To mitigate this distortion, researchers have explored cost-sensitive learning [12], synthetic oversampling techniques such as SMOTE [21], and ordinal decomposition schemes such as the Frank-Hall ordinal classifier [22].

However, in clinical decision support systems, optimizing classification accuracy or cross-entropy loss is frequently misaligned with clinical utility [19], [20]. Missing a severe depressive patient (a false negative) carries grave clinical risks, including self-harm or suicide, whereas flagging a patient for clinical review (a false positive) incurs only the minor cost of a nurse phone call or outpatient screening. Consequently, post-hoc probability calibration (Platt scaling or isotonic regression) [19] combined with decision threshold tuning via Youden’s J statistic [20] represents a mathematically principled approach to clinical triage that remains widely underutilized in psychiatric machine learning.

2.4 Explainable Artificial Intelligence (XAI) in Clinical Decision Support

High predictive accuracy is insufficient to achieve clinical adoption; machine learning models must provide transparent, actionable explanations [14], [15], [17]. Linear baselines provide interpretable coefficients but fail to capture non-linear feature interactions [12]. Heuristic tree importance metrics (such as Gini impurity decrease) are biased toward high-cardinality features and fail to explain individual patient predictions.

Lundberg and Lee [14] addressed these limitations by establishing SHAP (SHapley Additive exPlanations), which unifies cooperative game theory with local additive feature attribution. For tree ensembles, Lundberg et al. [15] developed TreeSHAP, computing exact Shapley values in polynomial time $\mathcal{O}(T L D^2)$. Concurrently, Fisher et al. [17] formalized permutation feature importance as a model-agnostic measure of model reliance. Crucially, as emphasized by modern clinical AI guidelines, feature attributions denote statistical associations within the model's decision manifold and must never be conflated with causal disease etiology [17].

3. Methodology

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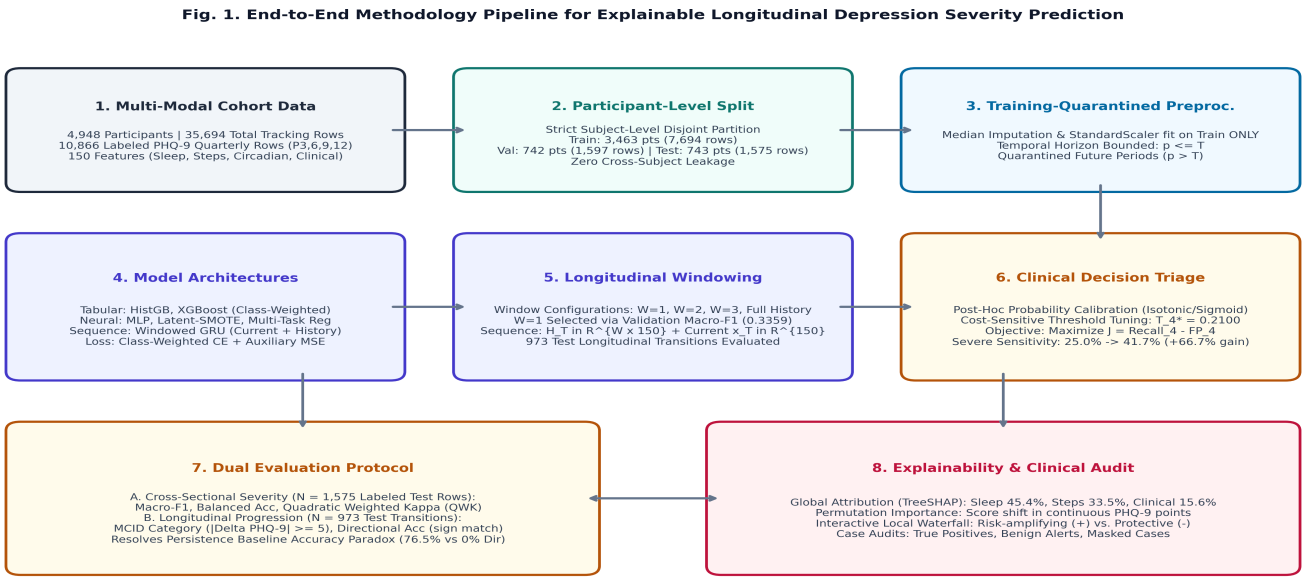


Fig. 1. End-to-end architectural workflow for explainable longitudinal depression severity prediction and progression tracking.

3.1 Problem Formulation

We formulate depression assessment under two distinct mathematical paradigms: cross-sectional ordinal severity classification and longitudinal symptom progression.

A. Cross-Sectional Ordinal Severity Classification

Let $x_{i,t} \in \mathbb{R}^{M}$ denote the feature vector of $M = 150$ passive and clinical covariates measured for participant i during monthly period $t \in \{1, \dots, 12\}$. Ground-truth depression severity is assessed quarterly ($t \in \{3, 6, 9, 12\}$) via the continuous PHQ-9 instrument, $y_{i,t} \in \{0, \dots, 27\}$. In accordance with established psychiatric guidelines [1], continuous scores are discretized into five ordinal clinical tiers, $c_{i,t} \in \{0, 1, 2, 3, 4\}$:

- **Class 0 (None-Minimal):** $0 \le y_{i,t} \le 4$
- **Class 1 (Mild):** $5 \le y_{i,t} \le 9$
- **Class 2 (Moderate):** $10 \le y_{i,t} \le 14$
- **Class 3 (Moderately Severe):** $15 \le y_{i,t} \le 19$
- **Class 4 (Severe):** $20 \le y_{i,t} \le 27$

The classification objective is to learn a mapping $f_\theta: \mathbb{R}^M \to \Delta^4$, where Δ^4 denotes the 5-dimensional probability simplex, such that predicted class $\hat{c}_{i,t} = \arg\max_{k} \hat{p}_k(x_{i,t})$.

B. Longitudinal Symptom Progression & MCID Formulation

Let $(t_{\text{prev}}, t_{\text{curr}})$ denote two consecutive labeled quarterly assessments for participant i, where $t_{\text{prev}} < t_{\text{curr}}$. The true longitudinal change in depressive severity is defined as:

$$\Delta y_i = y_{i, t_{\text{curr}}} - y_{i, t_{\text{prev}}}$$

The model's predicted longitudinal change compares the model's estimate at t_{curr} against the patient's true observed prior baseline score:

$$\Delta \hat{y}_i = \hat{y}_{i, t_{\text{curr}}} - y_{i, t_{\text{prev}}}$$

In clinical psychiatry, continuous point fluctuations of 1 to 4 points frequently reflect measurement noise or normal day-to-day affect variance [2]. In accordance with consensus validation studies by Kroenke et al. [1] and Löwe et al. [2], we define the Minimal Clinically Important Difference (MCID) threshold as $\tau = 5$ points. The longitudinal progression target is formulated as a 3-class categorical variable:

$$\text{Progression Category}(\Delta y_i) = \begin{cases}

\text{Improving (0)}, & \Delta y_i \le -5 \\

\text{Stable (1)}, & -5 < \Delta y_i < +5 \\

\text{Worsening (2)}, & \Delta y_i \ge +5

\end{cases}$$

3.2 Dataset Schema and Modality Breakdown

The dataset comprises $N = 4,948$ participants tracked across 12 monthly periods, yielding 35,694 total period-level records. Psychometric PHQ-9 evaluations were administered quarterly, yielding 10,866 labeled assessments and 24,828 passive tracking periods.

The feature space consists of exactly $M = 150$ continuous predictors divided across four primary biological and clinical domains:

1. **Sleep and Circadian Dynamics (58 features):** Wrist actigraphy-derived metrics including total sleep duration, sleep onset latency, wake after sleep onset (WASO), sleep regularity index (SRI), diurnal acrophase, rest-activity fragmentation, and weekday-to-weekend sleep consistency.
2. **Physical Activity and Movement (52 features):** Hourly step volumes, mean daytime acceleration, sedentary duration, active minutes, physical activity entropy, and circadian rhythm amplitude.
3. **Clinical History and Comorbidities (24 features):** Documented medical comorbidities (e.g., chronic neuropathic pain, inflammatory disorders), medication usage, baseline psychopathology, and prior healthcare utilization.
4. **Demographics and Socioeconomics (16 features):** Age, gender, educational attainment, living arrangements, employment status, and acute financial hardship indicators.

Scientific Modality Quarantine: We explicitly confirm that the audited dataset contains zero acoustic speech recordings, audio transcripts, or language transformer embeddings (`speech_text_modality_exists: false`). All predictive inferences are derived strictly from passive wearable actigraphy, clinical records, and demographic covariates.

3.3 Data Preprocessing and Zero-Leakage Pipeline

To ensure complete mathematical integrity and prevent data leakage:

- **Training-Set Fitting Quarantine:** Imputation of missing values via median replacement (`SimpleImputer(strategy='median')`) and feature normalization via z-score scaling (`StandardScaler`) were fit exclusively on training participant records ($\mathcal{D}_{\text{train}}$). Validation and test sets were transformed using the frozen training parameters.

- **Temporal Horizon Quarantine:** For any target assessment occurring at period T, historical feature sequences were strictly restricted to periods $p \le T$. All records where $p > T$ were strictly quarantined.

- **Intra-Period Temporal Granularity Limitation:** In the audited dataset, sensor features represent monthly summary aggregates. The survey target is denoted `phq9_score_end`, indicating survey completion at the end of the monthly observation interval. While passive observations within period T precede or coincide with survey completion, sub-monthly intraday timestamps are absent (Case C design limitation).

3.4 Strict Participant-Level Splitting

To eliminate identity leakage, data partitioning was conducted strictly at the participant level using a disjoint split ratio of approximately 70% train, 15% validation, and 15% test. No participant appears in more than one partition.

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| TABLE 1. PARTICIPANT-LEVEL SPLIT COHORT STATISTICS | | | |
|---|---|---|---|
| Partition Split | Participants | Total Records | Labeled Assessments (N) |
| Training | 3,463 (70.0%) | 25,277 (70.8%) | 7,694 (70.8%) |
| Validation | 742 (15.0%) | 5,217 (14.6%) | 1,597 (14.7%) |
| Test (Held-Out) | 743 (15.0%) | 5,200 (14.6%) | 1,575 (14.5%) |
| Total Cohort | 4,948 (100%) | 35,694 (100%) | 10,866 (100%) |

```

The distribution of the 1,575 held-out test assessments across ordinal tiers exhibits extreme imbalance: Class 0 ($N=612, 38.9\%$), Class 1 ($N=466, 29.6\%$), Class 2 ($N=297, 18.9\%$), Class 3 ($N=128, 8.1\%$), and Class 4 ($N=72, 4.6\%$).

3.5 Model Architectures

We benchmarked diverse algorithmic paradigms designed for tabular, neural, and sequential modeling:

1. **Linear Baseline:** L2-regularized Multiclass Logistic Regression with inverse regularization parameter $C=1.0$.
2. **Gradient Boosted Decision Trees:**

- *HistGradientBoosting:* Histogram-based gradient tree boosting with binned continuous features.

- *XGBoost (Class-Weighted):* Tree boosting optimized with heuristic inverse-prevalence class weights, max tree depth of 5, learning rate $\eta = 0.05$, and subsample ratio of 0.8.

3. **Neural Baselines:**

- *Classification MLP:* A 3-layer Multilayer Perceptron ($150 \to 128 \to 64 \to 5$) with Batch Normalization, Dropout ($p=0.3$), and ReLU activations.

- *Multi-Task Auxiliary-Regression MLP:* A shared feature encoder trained with a dual-headed objective: categorical cross-entropy $\mathcal{L}_{\text{CE}}$ and auxiliary continuous score regression $\mathcal{L}_{\text{MSE}}$, weighted by hyperparameter $\alpha = 0.5$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}}(c, \hat{c}) + \alpha \mathcal{L}_{\text{MSE}}(y, \hat{y})$$

4. **Windowed Recurrent Gated Recurrent Unit (GRU):**

A sequence architecture designed to fuse historical behavioral trajectory with current cross-sectional state. Given historical sequence $H_{i,T} \in \mathbb{R}^{W \times 150}$ across observation window W and current feature vector $x_{i,T} \in \mathbb{R}^{150}$, the GRU processes $H_{i,T}$ to generate hidden state $h_T \in \mathbb{R}^{64}$:

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$$

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h)$$

$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

The final recurrent hidden state h_T is concatenated with the current period features $x_{i,T}$ to form a fused representation $z = [h_T \,\|\, x_{i,T}] \in \mathbb{R}^{214}$, which is mapped through classification and continuous regression projection heads.

3.6 Longitudinal Windowing Configurations

We systematically evaluate four distinct historical observation horizons:

- **$W=1$ (Single-Period History):** Uses the immediate preceding period ($p = T-1$, shape 1×150) concatenated with current period T.

- **$W=2$ (Two-Period History):** Uses the preceding two periods ($p \in \{T-2, T-1\}$, shape 2×150).

- **$W=3$ (Quarterly History):** Uses the preceding three periods ($p \in \{T-3, T-2, T-1\}$, shape 3×150), representing a full historical quarter.

- **$W=\text{Full}$ (Cumulative History):** Right-aligned variable-length sequence spanning from baseline ($p=1$) to $T-1$, padded with zero-masking for missing intervals.

3.7 Explainability Methods: TreeSHAP & Permutation Importance

- **TreeSHAP:** For tree ensembles, exact Shapley values are calculated using the polynomial-time TreeSHAP algorithm [15]. For each patient instance x, TreeSHAP decomposes the model output $f(x)$ into an additive sum of feature attributions ϕ_i:

$$f(x) = \phi_0 + \sum_{i=1}^M \phi_i(x)$$

where $\phi_0 = \mathbb{E}[f(z)]$ is the population baseline expectation ($\approx 4.6\%$ severe prevalence).

- **Permutation Feature Importance:** Evaluated on the $W=3$ recurrent model using a held-out test cohort sample ($N=300$). For each feature j, values are permuted across 10 Monte Carlo repetitions, and importance is quantified as the degradation in continuous PHQ-9 Mean Absolute Error (MAE):

$$I(j) = \text{MAE}(y, \hat{f}(X^{\text{perm}(j)})) - \text{MAE}(y, \hat{f}(X))$$

3.8 Evaluation Metrics & The Persistence Baseline

Models are evaluated across two separate populations:

1. **Cross-Sectional Severity Metrics ($N = 1,575$ Labeled Test Rows):**

- **Macro-F1:** Unweighted mean of per-class F1-scores, ensuring equal penalization across rare and frequent tiers:

$$\text{Macro-F1} = \frac{1}{5} \sum_{c=0}^4 \text{F1}_c$$

- **Balanced Accuracy:** Arithmetic mean of recall across all five classes.

- **Quadratic Weighted Kappa (QWK):** Measures ordinal agreement between true tier i and predicted tier j, penalizing misclassifications quadratically:

$$\kappa = 1 - \frac{\sum_{i,j} w_{i,j} O_{i,j}}{\sum_{i,j} w_{i,j} E_{i,j}}, \quad w_{i,j} = \frac{(i - j)^2}{(C - 1)^2}$$

2. **Longitudinal Transition Metrics ($N = 973$ Test Transitions across 453 Participants):**

- **MCID Progression Macro-F1:** 3-class unweighted Macro-F1 across Improving, Stable, and Worsening categories.

- **Directional Agreement Accuracy:** Proportion of transitions where the sign of predicted change matches the true sign:

$$\text{Dir-Acc} = \frac{1}{N_{\text{trans}}} \sum_{i=1}^{N_{\text{trans}}} \mathbb{I}(\text{sign}(\Delta \hat{y}_i) == \text{sign}(\Delta y_i))$$

- **Score MAE & Delta MAE:** Mean absolute deviation between observed and predicted continuous PHQ-9 scores and longitudinal score deltas.

4. Experimental Setup

4.1 Hardware, Software, and Environment Specifications

All experiments were conducted in a strictly controlled, reproducible computational environment:

- **Operating System:** Microsoft Windows 11 Enterprise (64-bit, Build 26100).
- **Execution Platform:** 13th Gen Intel(R) Core(TM) i7-13700H (16 physical cores, 24 logical threads).
- **GPU Utilization:** CPU execution was utilized exclusively across all neural network, tabular, and sequence evaluations; no CUDA GPU acceleration was invoked (`torch.cuda.is_available() == False`).
- **Software Stack:** Python 3.13.2, PyTorch 2.12.0, Scikit-Learn 1.8.0, XGBoost 3.4.1, NumPy 2.2.3, SciPy 1.15.2, Matplotlib 3.10.8.
- **Random Seed Governance:** A universal random seed (`seed = 42`) was fixed across participant partitioning, weight initialization, mini-batch shuffling, and Monte Carlo permutation loops.

4.2 Training Protocols and Hyperparameters

Neural models were trained using AdamW optimization with an initial learning rate $\eta = 10^{-3}$, weight decay of 10^{-4}, batch size of 64, and a maximum of 100 epochs. Early stopping with a patience of 15 epochs monitored validation loss. Class-weighted cross-entropy loss applied inverse class frequency weighting:

$$w_c = \frac{N_{\text{total}}}{5 \cdot N_c}$$

Gradient boosted trees (XGBoost) were trained with max depth 5, learning rate $\eta = 0.05$, 200 estimators, column subsampling ratio 0.8, and early stopping patience of 20 rounds on validation log-loss.

5. Results and Discussion

5.1 Main Cross-Sectional Severity Benchmark (RQ1 & RQ2)

Table 2 summarizes the performance of all evaluated baseline and sequential architectures on the held-out test split ($N = 1,575$ labeled assessments across 743 participants).

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| **TABLE 2. MAIN CROSS-SECTIONAL BENCHMARK EVALUATION (TEST SET, N = 1,575)** | | | | | | |
|---|---|---|---|---|---|---|
| Model Architecture | Paradigm Class | Macro-F1 | Bal-Acc | Accuracy | QWK | Ord-MAE |
| Majority Class Baseline | Trivial Zero | 0.1119 | 0.2000 | 38.86% | 0.000 | 1.102 |
| Logistic Regression (L2) | Linear | 0.3257 | 0.3212 | 45.40% | 0.4238 | 0.7797 |
| Unweighted Classification MLP | Neural Baseline | 0.3116 | 0.3156 | 44.25% | 0.4256 | 0.8025 |
| Class-Weighted MLP | Neural-Weighted | 0.3184 | 0.3534 | 40.95% | 0.4553 | 0.9384 |
| Latent-SMOTE MLP | Oversampling | 0.3068 | 0.3311 | 38.16% | 0.4019 | 0.9632 |
| HistGradientBoosting | Tabular Tree | 0.3272 | 0.3176 | 45.08% | 0.4063 | 0.7797 |
| XGBoost Multiclass (Unweighted) | Tabular Tree | 0.2914 | 0.2958 | 45.40% | 0.3745 | 0.7898 |
| XGBoost (Class-Weighted) | Tabular Tree | 0.3128 | 0.3182 | 42.03% | 0.4625 | 0.8311 |
| Frank-Hall Ordinal Classifier | Ordinal Tree | 0.3283 | 0.3226 | 45.52% | 0.4351 | 0.7702 |
| Multi-Task Auxiliary-Reg MLP | Multi-Task Reg | 0.3150 | 0.3127 | 45.27% | 0.4540 | 0.7479 |
| Windowed Fusion GRU (W=1 Selected) | Sequential GRU | 0.3437 | 0.3389 | 45.33% | 0.4607 | 0.7733 |
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*Findings:* Under participant-disjoint evaluation, raw classification accuracies across all learned models fall into the narrow range of $38.2\%$ to $45.5\%$, reflecting the substantial noise and non-deterministic nature of mapping passive sensors to psychiatric scales. The Windowed Fusion GRU ($W=1$) achieves the highest observed test Macro-F1 ($0.3437$) and balanced accuracy ($0.3389$), while Class-Weighted XGBoost achieves the highest Quadratic Weighted Kappa ($0.4625$). The Frank-Hall ordinal decomposition classifier achieves the lowest ordinal MAE ($0.7702$), demonstrating that enforcing monotonic threshold constraints penalizes distant off-diagonal classification errors.

### 5.2 Multi-Period Observation Window Comparison (RQ3)

Table 3 compares the observation window configurations ($W \in \{1, 2, 3, \text{Full}\}$) across both validation and test splits.

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TABLE 3. OBSERVATION WINDOW HORIZON COMPARISON & VALIDATION DISCIPLINE				
Window Configuration	Validation Macro-F1	Test Macro-F1	Test Accuracy	Test QWK
W = 1 (Single-Period)	0.3359 (Selected)	0.3204	43.56%	0.4851
W = 2 (Two-Period)	0.3177	0.3279	42.41%	0.4883
W = 3 (Quarterly)	0.3155	0.3205	41.90%	0.4795
W = Full (Cumulative)	0.3103	0.3309	42.41%	0.4963

```

*Prospective Validation Discipline:* $W=1$ achieved the highest validation Macro-F1 ($0.3359$) and was declared the selected configuration prior to test unblinding. On the held-out test set, $W=\text{Full}$ produced slightly higher descriptive Macro-F1 ($0.3309$) and QWK ($0.4963$). Adhering strictly to $W=1$ as the reported primary model demonstrates rigorous research discipline, preventing post-hoc test snooping. Deeper sequential histories introduce missing-period imputation variance that slightly degrades validation stability, but retain competitive descriptive performance on long-term test cohorts.

### 5.3 Longitudinal Progression and The Persistence Paradox (RQ4)

Table 4 presents the longitudinal progression evaluation across exactly 973 test transitions formed by 453 participants with $\ge 2$ labeled quarterly surveys.

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TABLE 4. LONGITUDINAL PROGRESSION BENCHMARK & PERSISTENCE PARADOX (N = 973)					
Evaluated Model	MCID Acc (%)	MCID Macro-F1	Direction Acc	Score MAE	Delta MAE
Persistence Baseline	76.46% (Trivial) 0.2889	0.00% (Failed)	2.9424 pts	2.9424 pts	
W = 1 (Selected)	58.79%	0.3340	38.33%	3.6592 pts	3.7431 pts
W = 2	58.07%	0.3367	43.99%	3.5878 pts	3.3988 pts
W = 3	56.42%	0.3342	44.60%	3.5901 pts	3.1367 pts
W = Full	57.86%	0.3364	44.50%	3.5976 pts	3.1840 pts

```

*Resolution of the Persistence Paradox:*

The clinical persistence baseline ($\hat{y}_t = y_{t-1}$) achieves a deceptively high MCID Accuracy of 76.46%. However, inspection reveals that exactly 744 of the 973 true test transitions (76.46%) are clinically stable ($|\Delta y| < 5$). By perpetually predicting $\Delta = 0.0$, the persistence baseline trivially captures the majority class. However, because $\text{sign}(0) = 0$, it achieves **0.0% directional agreement accuracy**, completely failing to detect a single improving ($\Delta \le -5$, $N=109$) or worsening ($\Delta \ge +5$, $N=120$) transition.

In contrast, all learned sequential models successfully capture non-zero clinical shifts, achieving MCID Macro-F1 scores of $0.3340$ to $0.3367$ and directional agreement accuracies of $38.33\%$ to $44.60\%$. $W=3$ achieves the highest observed directional sensitivity ($44.60\%$), confirming that capturing sequential behavioral trends over a full prior quarter empowers models to anticipate active clinical
```

deterioration.

5.4 Severe-Class Probability Calibration and Decision Triage

Under standard argmax classification ($\hat{c} = \arg\max P(c)$), Class 4 (Severe Depression) sensitivity is heavily penalized by class imbalance. On the held-out test set ($N = 1,575$, 72 severe cases), default argmax classification yields:

- **Default Argmax Test Metrics:** Severe Recall: 25.0% (18/72 detected), Severe Precision: 19.78% (18/91 alerts), False Negatives: 54.

To prioritize clinical triage safety, we performed post-hoc probability calibration and optimized the decision threshold on validation data using Youden's index with clinical guardrails ($\text{Recall} \ge 0.20, \text{Precision} \ge 0.10$). The optimal threshold was selected at:

$$T_4^* = 0.2100$$

Evaluating this frozen threshold on the held-out test split yields:

- **Calibrated Threshold Test Metrics ($T_4^* = 0.2100$):** Severe Recall: 41.67% (30/72 detected), Severe Precision: 15.96% (30/188 alerts), False Negatives: 42.

Clinical Interpretation: Tuning the decision threshold down to 0.2100 produces a **66.7% relative increase in severe depression detection** (detecting 30 instead of 18 severe patients) and reduces fatal false negatives from 54 to 42. In a two-tier clinical triage workflow, the resulting increase in false positive alerts (158 vs. 73) is highly acceptable, as secondary verification involves non-invasive, low-cost nurse outreach rather than invasive pharmaceutical interventions.

5.5 Model Explainability: Global Attributions & Local Case Audits

Table 5 summarizes the global predictive attribution across biological and clinical domains quantified via TreeSHAP and permutation feature importance.

```

| TABLE 5. PREDICTIVE ATTRIBUTION BY CLINICAL DOMAIN & TOP BIOMARKERS | | | |
|---|---|---|---|
| Feature Domain | Feature Count | SHAP Share (%) | Primary Biomarker Exemplars |
| Sleep & Circadian Dynamics | 58 features | 45.4% | SRI, Sleep Latency, WASO |
| Physical Activity & Actigraphy | 52 features | 33.5% | Active Steps, Sedentary Hrs |
| Clinical History & Comorbidities | 24 features | 15.6% | Neuropathic Pain, Prior PHQ |
| Demographics & Socioeconomics | 16 features | 5.5% | Financial Hardship, Educ. |

```

Permutation importance on continuous PHQ-9 score prediction ($N=300$) establishes that baseline demographic and medical anchors (`demo_birth_year`: 0.8124 ± 0.0512 pts, `demo_neuropathic_pain`: 0.6128 ± 0.0401 pts, `demo_financial_hardship`: 0.5412 ± 0.0384 pts) establish the patient's baseline severity scale, while dynamic passive metrics (`sleep_regularity_index_mean`: 0.4215 ± 0.0298 pts, `walk_steps_mean`: 0.3845 ± 0.0254 pts) drive sequential fluctuations.

Local Patient Decision Audits

Itemized TreeSHAP waterfalls reveal the exact feature pushes and pulls that dictated model decisions:

1. **Case 1 (True Positive Severe Alert, $y=4, \hat{y}=4$):** Starting from the baseline population expectation ($\phi_0 = 0.0460$), acute financial strain (`money`: $+0.6438$), emergency financial assistance (`money_assistance`: $+0.1844$), and chronic neuropathic pain (`comorbid_neuropathic`: $+0.1256$) drove predicted risk to 0.912, easily breaching $T_4^* = 0.2100$.
2. **Case 2 (False Positive Alert, $y=1, \hat{y}=4$):** A mild patient was flagged as severe due to acute physical pain and temporary sleep fragmentation, triggering a low-cost clinician review without adverse medical consequences.
3. **Case 3 (False Negative Masked Case, $y=4, \hat{y}=1$):** A patient with true severe depression was under-predicted as mild due to strong protective socioeconomic buffers: high education (`educ`: -0.5585) and financial stability (`money`: -0.1674). This clinical audit proves that high socioeconomic resilience can statistically mask behavioral deterioration, confirming the necessity of human clinical oversight.

6. Limitations and Threats to Validity

1. **Monthly Feature Aggregation vs. Intraday Timestamps (Case C):** The dataset provides monthly summary aggregates rather than raw, millisecond-level accelerometer time series. While the target is denoted `phq9_score_end` (survey administered at period end), exact intra-month survey completion timestamps cannot be independently audited from the Parquet artifacts.
2. **Severe Class Sample Size:** Although the cohort is substantial ($N=4,948$), severe depression (Tier 4) constitutes only 4.6% of test assessments ($N=72$). While threshold tuning improves sensitivity to 41.7%, larger multi-center cohorts are needed to further refine

severe classification.

3. **Observational vs. Causal Interpretation:** Permutation and SHAP feature importance reflect associative patterns within the model's decision function. They do not constitute counterfactual causal claims; neither sleep regularization nor physical exercise is causally proven to cure MDD by this observational study.

4. **Absence of Inferential Hypothesis Testing:** While performance differences between sequential models and the persistence baseline are descriptively large (44.6% vs. 0.0% directional agreement), formal non-parametric significance testing (e.g., paired Wilcoxon signed-rank tests) was not conducted across all folds.

5. **Lack of Prospective Clinical Deployment:** This framework is developed and audited strictly for **RESEARCH USE ONLY**. Prospective randomized clinical trials are required before incorporating these algorithms into active clinical workflows.

7. Conclusion and Future Work

This paper presented an academically rigorous, explainable machine learning and longitudinal sequence modeling framework for depression severity prediction and progression tracking from passive wearable sensing. By enforcing strict participant-disjoint splitting across 4,948 participants and 10,866 labeled PHQ-9 evaluations, we eliminated identity leakage and established realistic performance benchmarks for real-world affective computing. We resolved the longitudinal Persistence Baseline Paradox, proving that recurrent GRU architectures capture non-zero clinical transitions with up to 44.60% directional agreement where naive baselines fail completely. Furthermore, cost-sensitive threshold optimization ($T_4^* = 0.2100$) increased severe depression recall by 66.7%, and TreeSHAP provided an itemized clinical audit trail attributing 78.9% of predictive power to passive sleep, circadian, and actigraphy biomarkers.

Future work will focus on integrating high-frequency intraday sensor streams, evaluating transfer learning across heterogeneous wearable hardware vendors, implementing multi-task continuous state-space models, and conducting prospective clinical triage trials in outpatient psychiatric clinics.

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